PHYS401: Quantum Mechanics I
Homework #10

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Problem 1: Infinite Spherical Well ($\ell = 0$)

Solution

(a) Normalized eigenstates and eigenvalues for $\ell = 0$

The potential is

$$V(r) = \begin{cases} 0, & 0 \le r < a, \\ \infty, & r \ge a. \end{cases}$$

For $\ell = 0$ (an s–state), the wavefunction separates as

$$\psi(r, \theta, \phi) = R(r)\, Y_{00}(\theta, \phi), \qquad Y_{00} = \frac{1}{\sqrt{4\pi}}.$$

Define the reduced radial function

$$u(r) \equiv r\, R(r).$$

For $\ell = 0$ and $0 < r < a$ (where $V = 0$), the radial Schrödinger equation reduces to

$$\frac{d^2 u}{dr^2} + k^2 u = 0, \qquad k^2 = \frac{2\mu E}{\hbar^2}.$$

Regularity at $r = 0$ requires $u(0) = 0$, and the infinite wall at $r = a$ requires

$$\psi(a) = 0 \;\Rightarrow\; R(a) = 0 \;\Rightarrow\; u(a) = 0.$$

The general solution is $u(r) = A\sin(kr) + B\cos(kr)$. The condition $u(0) = 0$ forces $B = 0$, so $u(r) = A\sin(kr)$. The condition $u(a) = 0$ gives

$$\sin(ka) = 0 \;\Rightarrow\; ka = n\pi, \qquad n = 1, 2, 3, \ldots$$

Hence

$$k_n = \frac{n\pi}{a}, \qquad E_n = \frac{\hbar^2 k_n^2}{2\mu} = \frac{\hbar^2 \pi^2}{2\mu a^2}\, n^2.$$

Using the normalization identity for $\ell = 0$,

$$\int_0^a |u(r)|^2\, dr = 1,$$

we obtain

$$1 = A^2 \int_0^a \sin^2\Big(\frac{n\pi r}{a}\Big)\, dr = A^2 \frac{a}{2} \;\Rightarrow\; A = \sqrt{\frac{2}{a}}.$$

Therefore the normalized reduced radial eigenfunctions are

$$\boxed{u_n(r) = \sqrt{\frac{2}{a}}\, \sin\Big(\frac{n\pi r}{a}\Big) \quad (0 < r < a), \qquad u_n(r) = 0 \;(r \geq a)}$$

and

$$\boxed{R_n(r) = \frac{u_n(r)}{r} = \sqrt{\frac{2}{a}}\, \frac{\sin\big(\frac{n\pi r}{a}\big)}{r} \quad (0 < r < a), \qquad R_n(r) = 0 \;(r \geq a).}$$

The full (normalized) wavefunctions are

$$\boxed{\psi_{n00}(r,\theta,\phi) = R_n(r)\, Y_{00}(\theta,\phi) = \frac{1}{\sqrt{4\pi}} \sqrt{\frac{2}{a}}\, \frac{\sin\big(\frac{n\pi r}{a}\big)}{r}, \quad (r < a), \qquad \psi_{n00} = 0 \;(r \geq a).}$$

(b) Plots of $P(r) = |u(r)|^2$ and $|R(r)|^2$ for the two lowest s–states

For $\ell = 0$, the radial probability in a shell $[r, r + dr]$ is

$$\Pr(r \in [r, r + dr]) = |u(r)|^2\, dr,$$

so $P(r) = |u(r)|^2$ is the physically relevant radial probability density in r. The quantity $|R(r)|^2$ is shown as well for comparison. On a log plot, true zeros cannot be displayed; in the $|R(r)|^2$ plot a small display floor is used so that zeros at nodes and at the wall appear consistently.

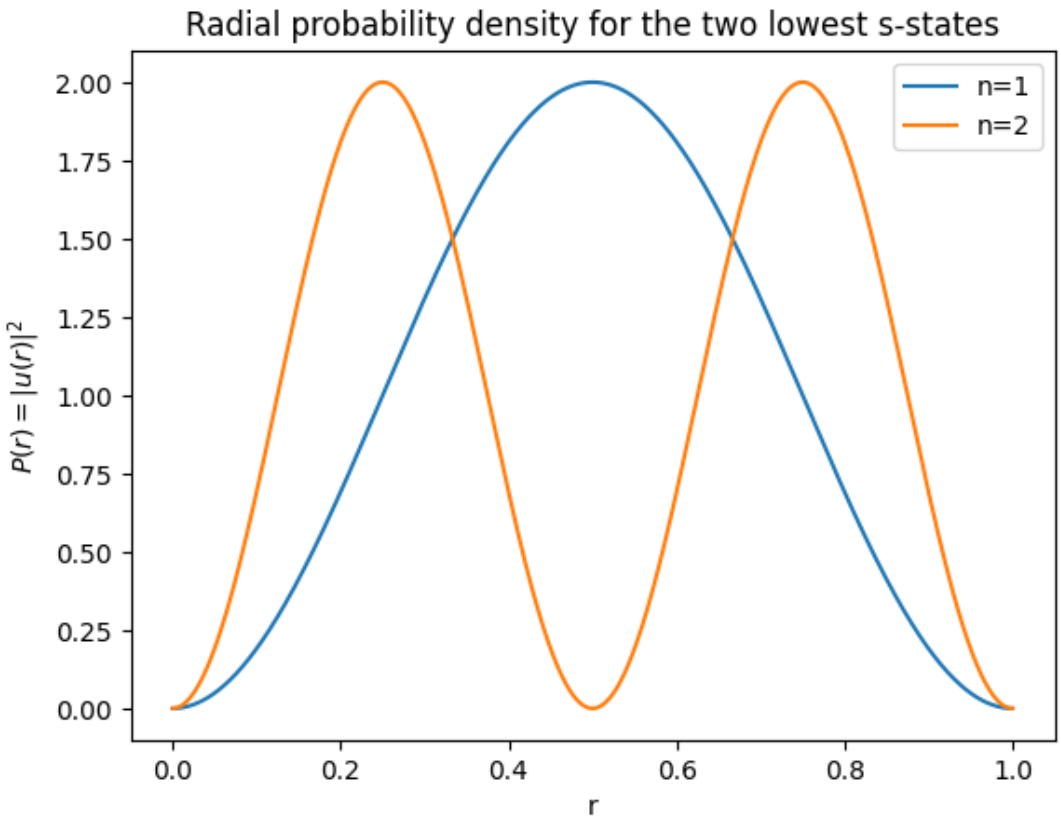

Figure 1: Infinite spherical well ($\ell = 0$): radial probability density $P(r) = |u(r)|^2$ for the two lowest states ($n = 1, 2$).

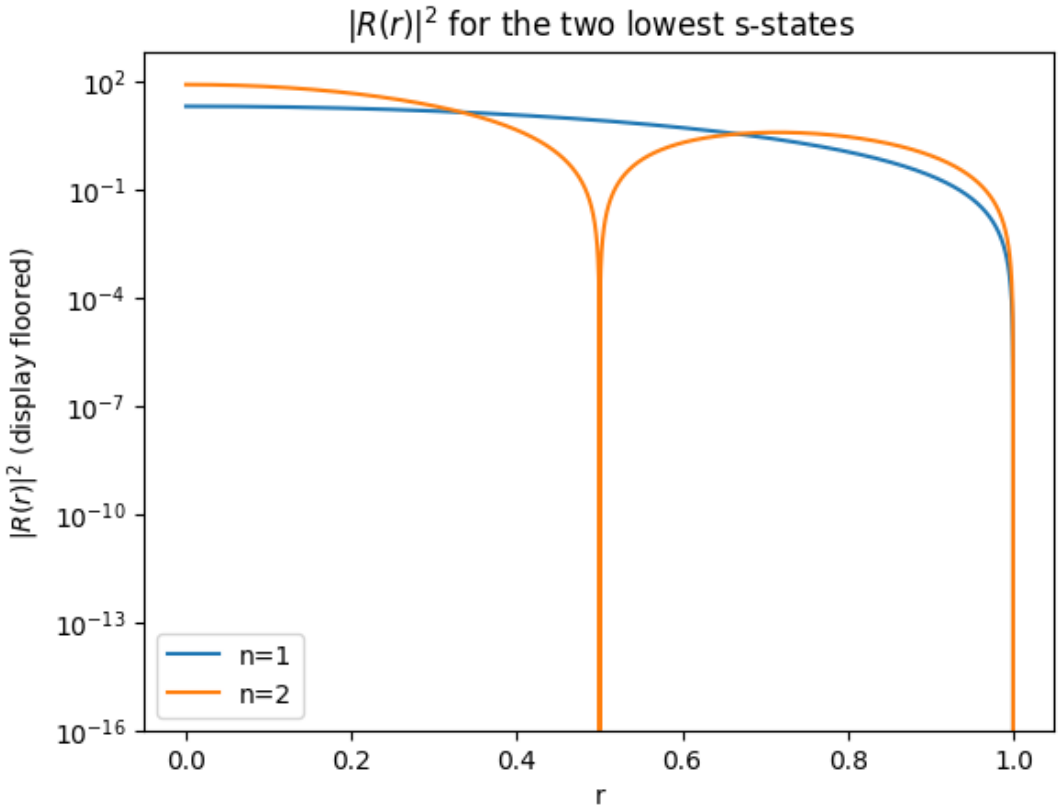

Figure 2: Infinite spherical well ($\ell = 0$): $|R(r)|^2$ for the two lowest states ($n = 1, 2$) on a log scale. Zeros at the node ($n = 2$ at $r = a/2$) and at the wall ($r = a$) are displayed with a common floor for readability.

(c) How this differs from hydrogen s–states

Although both problems involve $\ell = 0$ (s–states), the physics and boundary conditions are different:

- **Potential:** the spherical well has a hard wall at $r = a$, whereas hydrogen has a Coulomb potential

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0 r}$$

  with no finite boundary.

- **Boundary condition at large r:** in the well, $u(a) = 0$ forces the wavefunction to vanish at $r = a$ exactly. In hydrogen, there is no wall; bound states satisfy normalizability, so the wavefunction decays smoothly as $r \to \infty$.

- **Spectrum:** the well has $E_n \propto n^2$ (standing–wave quantization from the boundary), while hydrogen has negative bound energies scaling like $E_n \propto -1/n^2$.

- **Radial shapes:** in the well, $u_n(r)$ is sinusoidal. In hydrogen, the s–state radial functions have an exponential factor times a polynomial, producing nodes (e.g. the $2s$ node) without any hard wall.

For reference, the analogous plots for the hydrogen $1s$ and $2s$ states are shown below.

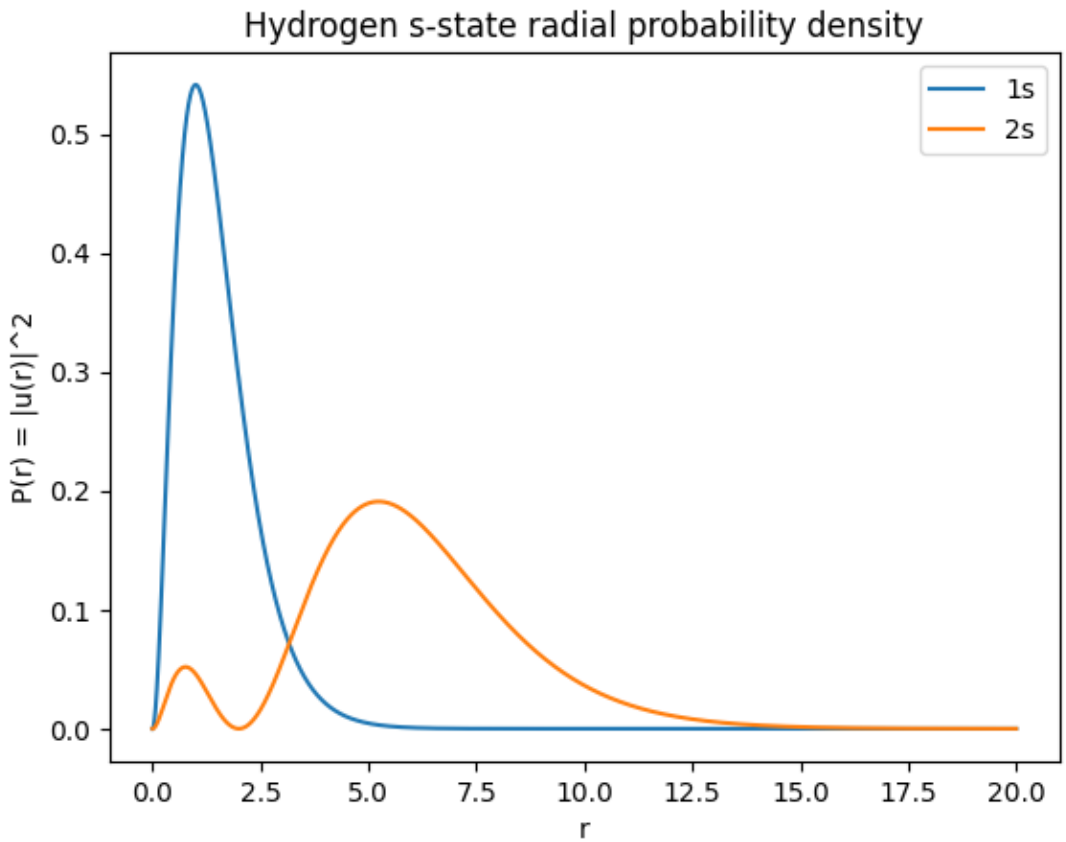

Figure 3: Hydrogen atom: radial probability density $P(r) = |u(r)|^2$ for the $1s$ and $2s$ states. Unlike the spherical well, the distribution is not forced to vanish at a finite radius; it decays with an exponential tail.

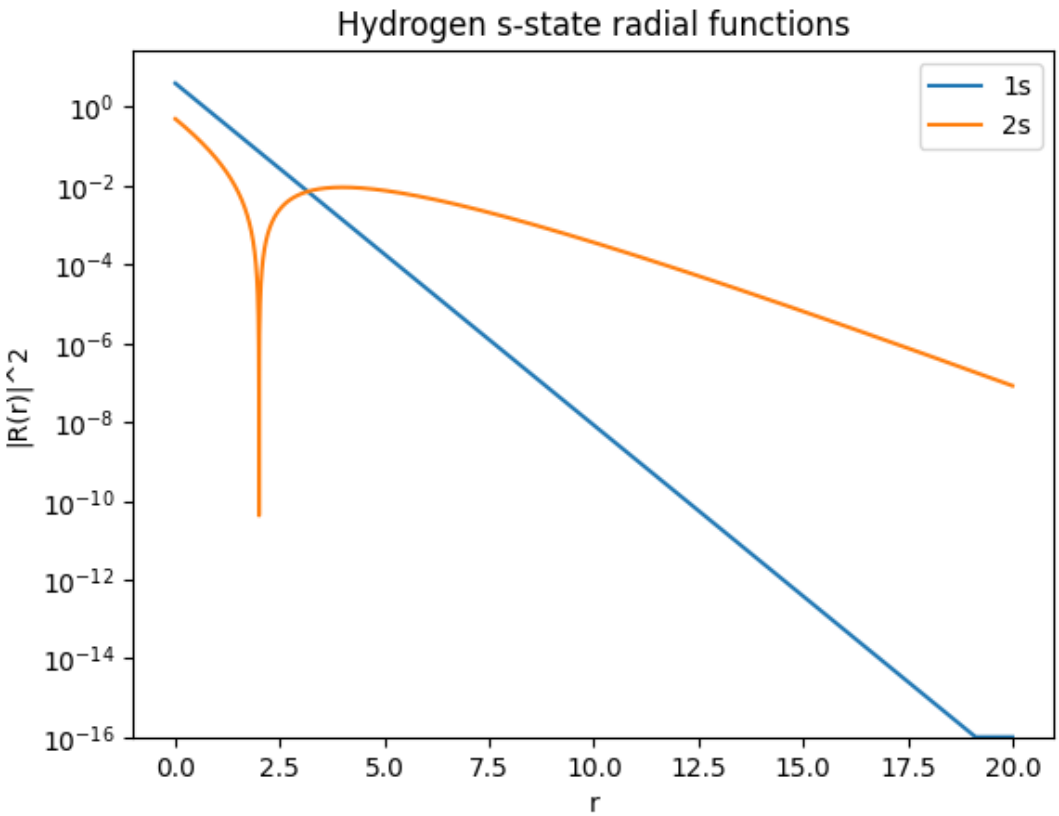

Figure 4: Hydrogen atom: $|R(r)|^2$ for the $1s$ and $2s$ states on a log scale. The $2s$ node appears at a finite r, but there is no hard-wall boundary forcing $R(r)$ to vanish at a fixed radius.

Problem 2: Sep. of Variables in Rect. Coords

Solution

(a) Derive the eigenstates and eigenvalues via separation of variables

Inside the box ($0 < x < a$, $0 < y < b$, $0 < z < c$) the potential is $V = 0$, so the time–independent Schrödinger equation (McIntyre Eq. 7.29) becomes

$$-\frac{\hbar^2}{2m}\nabla^2\psi(x,y,z) = E\,\psi(x,y,z), \qquad \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

Equivalently,

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{2mE}{\hbar^2}\psi = 0. \tag{1}$$

We seek separated solutions of the form

$$\psi(x,y,z) = X(x)\,Y(y)\,Z(z).$$

Substituting into Eq. (1) gives

$$X''YZ + XY''Z + XYZ'' + \frac{2mE}{\hbar^2}XYZ = 0.$$

Divide through by XYZ:

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} + \frac{2mE}{\hbar^2} = 0. \tag{2}$$

The first term depends only on x, the second only on y, and the third only on z, so each must be equal to a constant. Choose separation constants $k_x^2, k_y^2, k_z^2 > 0$ such that

$$\frac{X''}{X} = -k_x^2, \qquad \frac{Y''}{Y} = -k_y^2, \qquad \frac{Z''}{Z} = -k_z^2,$$

and therefore Eq. (2) implies

$$k_x^2 + k_y^2 + k_z^2 = \frac{2mE}{\hbar^2}. \tag{3}$$

This yields three ordinary differential equations:

$$X'' + k_x^2 X = 0, \qquad Y'' + k_y^2 Y = 0, \qquad Z'' + k_z^2 Z = 0. \tag{4}$$

These are exactly the same form as the 1D infinite square well (Ch. 5).

Because the potential is infinite outside the region, the wavefunction must vanish on each boundary:

$$\psi(0,y,z) = \psi(a,y,z) = 0, \quad \psi(x,0,z) = \psi(x,b,z) = 0, \quad \psi(x,y,0) = \psi(x,y,c) = 0.$$

For separated solutions this implies

$$X(0) = X(a) = 0, \qquad Y(0) = Y(b) = 0, \qquad Z(0) = Z(c) = 0. \tag{5}$$

Solving $X'' + k_x^2 X = 0$ gives $X = A\sin(k_x x) + B\cos(k_x x)$. The condition $X(0) = 0$ forces $B = 0$, and $X(a) = 0$ requires

$$\sin(k_x a) = 0 \;\Rightarrow\; k_x a = n_x\pi, \qquad n_x = 1, 2, 3, \ldots$$

so

$$k_x = \frac{n_x \pi}{a}, \qquad X_{n_x}(x) = A \sin\left(\frac{n_x \pi x}{a}\right).$$
$$k_y = \frac{n_y \pi}{b}, \qquad Y_{n_y}(y) = C \sin\left(\frac{n_y \pi y}{b}\right),$$
$$k_z = \frac{n_z \pi}{c}, \qquad Z_{n_z}(z) = D \sin\left(\frac{n_z \pi z}{c}\right),$$

with $n_y, n_z = 1, 2, 3, \ldots$

Using Eq. (3), the energy is

$$E = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right) = \frac{\hbar^2}{2m}\left(\frac{n_x^2 \pi^2}{a^2} + \frac{n_y^2 \pi^2}{b^2} + \frac{n_z^2 \pi^2}{c^2}\right).$$

Thus,

$$\boxed{E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2m}\left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2}\right)} \qquad (n_x, n_y, n_z \in \mathbb{Z}^+).$$

Finally, normalize the eigenfunction. Because the variables separate,

$$1 = \int_0^a \int_0^b \int_0^c |\psi|^2 \, dx \, dy \, dz = \left(\int_0^a |X|^2 dx\right)\left(\int_0^b |Y|^2 dy\right)\left(\int_0^c |Z|^2 dz\right).$$

Using the standard 1D result

$$\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{a}{2},$$

we choose $A = \sqrt{2/a}$, $C = \sqrt{2/b}$, $D = \sqrt{2/c}$. Therefore the normalized stationary states are

$$\boxed{\phi_{n_x n_y n_z}(x, y, z) = \sqrt{\frac{2}{a}}\sqrt{\frac{2}{b}}\sqrt{\frac{2}{c}} \, \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right)}$$

for $0 < x < a$, $0 < y < b$, $0 < z < c$, and $\phi = 0$ otherwise.

(b) Lowest 4 distinct energy levels for $b = c = \sqrt{2}\, a$

From part (a),

$$E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2m}\left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2}\right).$$

With

$$b = \sqrt{2}\, a, \qquad c = \sqrt{2}\, a,$$

we have $b^2 = c^2 = 2a^2$, so

$$E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2ma^2}\left(n_x^2 + \frac{n_y^2}{2} + \frac{n_z^2}{2}\right) = \frac{\hbar^2 \pi^2}{4ma^2} \underbrace{\left(2n_x^2 + n_y^2 + n_z^2\right)}_{\equiv Q}. \tag{6}$$

Thus distinct energies correspond to distinct integer values of

$$Q = 2n_x^2 + n_y^2 + n_z^2, \qquad n_x, n_y, n_z = 1, 2, 3, \ldots$$

We now list the smallest values of Q and the corresponding quantum numbers.

- **Ground level:** $(n_x, n_y, n_z) = (1,1,1)$ gives

$$Q = 2(1)^2 + 1^2 + 1^2 = 4 \quad \Rightarrow \quad E_1 = \frac{\hbar^2\pi^2}{4ma^2}(4) = \frac{\hbar^2\pi^2}{ma^2}.$$

- **Second distinct level:** $(1,1,2)$ and $(1,2,1)$ give

$$Q = 2(1)^2 + 1^2 + 2^2 = 7 \quad \Rightarrow \quad E_2 = \frac{\hbar^2\pi^2}{4ma^2}(7) = \frac{7}{4}\frac{\hbar^2\pi^2}{ma^2}.$$

- **Third distinct level:** $(1,2,2)$ and $(2,1,1)$ give

$$Q = 2(1)^2 + 2^2 + 2^2 = 10, \qquad Q = 2(2)^2 + 1^2 + 1^2 = 10,$$

  so both correspond to

$$E_3 = \frac{\hbar^2\pi^2}{4ma^2}(10) = \frac{5}{2}\frac{\hbar^2\pi^2}{ma^2}.$$

- **Fourth distinct level:** $(1,1,3)$ and $(1,3,1)$ give

$$Q = 2(1)^2 + 1^2 + 3^2 = 12 \quad \Rightarrow \quad E_4 = \frac{\hbar^2\pi^2}{4ma^2}(12) = 3\,\frac{\hbar^2\pi^2}{ma^2}.$$

(c) Degeneracies of the levels

Because $b = c$, the Hamiltonian is symmetric under interchange $y \leftrightarrow z$. Therefore, whenever $n_y \neq n_z$, the two states (n_x, n_y, n_z) and (n_x, n_z, n_y) have the same energy, giving at least a twofold degeneracy. In addition, there can be "accidental" degeneracy when different triples produce the same value of $Q = 2n_x^2 + n_y^2 + n_z^2$.

For the four lowest distinct levels found above:

- E_1 (from $(1,1,1)$): degeneracy $g_1 = 1$.
- E_2 (from $(1,1,2)$ and $(1,2,1)$): degeneracy $g_2 = 2$ (swap y and z).
- E_3 (from $(1,2,2)$ and $(2,1,1)$): degeneracy $g_3 = 2$ (accidental degeneracy).
- E_4 (from $(1,1,3)$ and $(1,3,1)$): degeneracy $g_4 = 2$ (swap y and z).

Problem 3: Angular Momentum Kets Practice

Solution

We are given the state at $t = 0$,

$$|\psi(0)\rangle = \frac{1}{\sqrt{12}}|0,0\rangle + \frac{\sqrt{2}}{\sqrt{12}}|1,-1\rangle + \frac{3}{\sqrt{12}}|1,0\rangle,$$

where $|\ell, m\rangle$ are simultaneous eigenstates of L^2 and L_z:

$$L_z|\ell,m\rangle = \hbar m|\ell,m\rangle, \qquad L^2|\ell,m\rangle = \hbar^2\ell(\ell+1)|\ell,m\rangle.$$

The state is normalized since

$$\frac{1}{12} + \frac{2}{12} + \frac{9}{12} = 1.$$

(a) Measurement of L_z

A measurement of L_z yields eigenvalues $\hbar m$ corresponding to the m values present in the state.

- $m = 0$ arises from both $|0,0\rangle$ and $|1,0\rangle$:

$$P(L_z = 0) = \left|\frac{1}{\sqrt{12}}\right|^2 + \left|\frac{3}{\sqrt{12}}\right|^2 = \frac{1}{12} + \frac{9}{12} = \frac{5}{6}.$$

- $m = -1$ arises from $|1,-1\rangle$:

$$P(L_z = -\hbar) = \left|\frac{\sqrt{2}}{\sqrt{12}}\right|^2 = \frac{2}{12} = \frac{1}{6}.$$

  There is no $|1,+1\rangle$ component, so $P(L_z = +\hbar) = 0$.

(b) Measurement of L^2

A measurement of L^2 yields eigenvalues $\hbar^2\ell(\ell+1)$.

- $\ell = 0 \Rightarrow L^2 = 0$:

$$P(L^2 = 0) = \left|\frac{1}{\sqrt{12}}\right|^2 = \frac{1}{12}.$$

- $\ell = 1 \Rightarrow L^2 = 2\hbar^2$:

$$P(L^2 = 2\hbar^2) = \left|\frac{\sqrt{2}}{\sqrt{12}}\right|^2 + \left|\frac{3}{\sqrt{12}}\right|^2 = \frac{2}{12} + \frac{9}{12} = \frac{11}{12}.$$

(c) Measure $L_z = -\hbar$, then measure L_x (using McIntyre spin–1 results)

If the measurement of L_z yields $-\hbar$, the state collapses onto the corresponding L_z eigenket in the $\ell = 1$ subspace:

$$|\psi\rangle \longrightarrow |1, -1\rangle .$$

For $\ell = 1$ (a spin–1 system), the possible outcomes of measuring L_x are

$$L_x = \hbar m_x, \qquad m_x \in \{+1, 0, -1\}.$$

McIntyre (spin–1 system) gives the S_x eigenkets written in the S_z basis as:

$$|+1\rangle_x = \frac{1}{2}|+1\rangle + \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{2}|-1\rangle ,$$

$$|0\rangle_x = \frac{1}{\sqrt{2}}|+1\rangle - \frac{1}{\sqrt{2}}|-1\rangle ,$$

$$|-1\rangle_x = \frac{1}{2}|+1\rangle - \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{2}|-1\rangle .$$

Here we identify $|+1\rangle \equiv |1, +1\rangle$, $|0\rangle \equiv |1, 0\rangle$, and $|-1\rangle \equiv |1, -1\rangle$ (McIntyre's notation).

We want the probabilities for outcomes of L_x when the system is in the state $|-1\rangle$. Using the projection postulate,

$$P(m_x) = |\langle m_{xx}| - 1\rangle|^2.$$

Compute the overlaps by taking inner products with $|-1\rangle$:

$$\langle +1_x| - 1\rangle = \left(\frac{1}{2}\langle +1| + \frac{1}{\sqrt{2}}\langle 0| + \frac{1}{2}\langle -1|\right)|-1\rangle = \frac{1}{2}\langle -1| - 1\rangle = \frac{1}{2},$$

$$\langle 0_x| - 1\rangle = \left(\frac{1}{\sqrt{2}}\langle +1| - \frac{1}{\sqrt{2}}\langle -1|\right)|-1\rangle = -\frac{1}{\sqrt{2}}\langle -1| - 1\rangle = -\frac{1}{\sqrt{2}},$$

$$\langle -1_x| - 1\rangle = \left(\frac{1}{2}\langle +1| - \frac{1}{\sqrt{2}}\langle 0| + \frac{1}{2}\langle -1|\right)|-1\rangle = \frac{1}{2}\langle -1| - 1\rangle = \frac{1}{2}.$$

Therefore the probabilities are

$$P(m_x = +1) = \left|\frac{1}{2}\right|^2 = \frac{1}{4}, \qquad P(m_x = 0) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}, \qquad P(m_x = -1) = \left|\frac{1}{2}\right|^2 = \frac{1}{4}.$$

Thus the possible measurement results for L_x are

$$\boxed{L_x = +\hbar\ (1/4), \qquad L_x = 0\ (1/2), \qquad L_x = -\hbar\ (1/4).}$$

(d) Measure $L^2 = 2\hbar^2$

(i) New state after the measurement

Measuring L^2 and obtaining $2\hbar^2$ projects the state onto the $\ell = 1$ subspace. The unnormalized projected state is

$$|\psi_{\ell=1}\rangle = \frac{\sqrt{2}}{\sqrt{12}}\,|1,-1\rangle + \frac{3}{\sqrt{12}}\,|1,0\rangle\,.$$

Its norm squared is

$$\frac{2}{12} + \frac{9}{12} = \frac{11}{12},$$

so the normalized post–measurement state is

$$\boxed{|\psi'\rangle = \sqrt{\frac{2}{11}}\,|1,-1\rangle + \frac{3}{\sqrt{11}}\,|1,0\rangle}.$$

(ii) Measure L_z after measuring L^2

In the state $|\psi'\rangle$, the possible L_z outcomes are

$$L_z = -\hbar \quad \text{and} \quad L_z = 0.$$

The probabilities are

$$P(L_z = -\hbar) = \left|\sqrt{\frac{2}{11}}\right|^2 = \frac{2}{11}, \qquad P(L_z = 0) = \left|\frac{3}{\sqrt{11}}\right|^2 = \frac{9}{11}.$$

There is no $m = +1$ component, so $P(L_z = +\hbar) = 0$.

Problem 4: Particle on a Ring

Solution

A particle of mass μ constrained to a ring of radius a has

$$\hat{H} = \frac{\hat{L}_z^2}{2\mu a^2}.$$

In lecture, simultaneous eigenkets of $\hat{H}$ and $\hat{L}_z$ were denoted $|m\rangle$, with

$$\hat{L}_z |m\rangle = \hbar m |m\rangle, \qquad \hat{H} |m\rangle = E_m |m\rangle, \qquad E_m = \frac{\hbar^2 m^2}{2\mu a^2}.$$

The initial state is

$$|\psi(0)\rangle = \frac{2}{5} |-3\rangle + \frac{2\sqrt{3}}{5} |+1\rangle + \frac{3}{5} |+3\rangle.$$

It is normalized since

$$\left|\frac{2}{5}\right|^2 + \left|\frac{2\sqrt{3}}{5}\right|^2 + \left|\frac{3}{5}\right|^2 = \frac{4}{25} + \frac{12}{25} + \frac{9}{25} = 1.$$

(a) Measure L_z

The possible results are the eigenvalues $\hbar m$ appearing in the expansion.

$$P(L_z = -3\hbar) = \left|\frac{2}{5}\right|^2 = \frac{4}{25}, \qquad P(L_z = +\hbar) = \left|\frac{2\sqrt{3}}{5}\right|^2 = \frac{12}{25}, \qquad P(L_z = +3\hbar) = \left|\frac{3}{5}\right|^2 = \frac{9}{25}.$$

(b) Measure energy

Since $E_m = \hbar^2 m^2/(2\mu a^2)$, the distinct energies present are:

- For $m = +1$:

$$E_1 = \frac{\hbar^2}{2\mu a^2}, \qquad P(E_1) = \frac{12}{25}.$$

- For $m = \pm 3$:

$$E_3 = \frac{9\hbar^2}{2\mu a^2}, \qquad P(E_3) = P(m = -3) + P(m = +3) = \frac{4}{25} + \frac{9}{25} = \frac{13}{25}.$$

(c) Compute $\langle \hat{L}_z \rangle$ and $\langle \hat{H} \rangle$

Using $\langle \hat{L}_z \rangle = \sum_m |c_m|^2 (\hbar m)$:

$$\langle \hat{L}_z \rangle = \hbar \left[(-3)\frac{4}{25} + (1)\frac{12}{25} + (3)\frac{9}{25} \right] = \hbar \left(\frac{-12 + 12 + 27}{25} \right) = \boxed{\frac{27}{25}\hbar}.$$

Using $\langle \hat{H} \rangle = \sum_m |c_m|^2 \, \hbar^2 m^2/(2\mu a^2)$:

$$\langle \hat{H} \rangle = \frac{\hbar^2}{2\mu a^2} \left[9\frac{4}{25} + 1\frac{12}{25} + 9\frac{9}{25} \right] = \frac{\hbar^2}{2\mu a^2} \left(\frac{36 + 12 + 81}{25} \right) = \boxed{\frac{129}{25} \frac{\hbar^2}{2\mu a^2}}.$$

(d) First measure $L_z = +\hbar$, then measure energy

If the first measurement yields $L_z = +\hbar$, then the state collapses to the corresponding eigenket:

$$|\psi\rangle \longrightarrow |+1\rangle .$$

But $|+1\rangle$ is also an energy eigenstate (since $\hat{H}$ and $\hat{L}_z$ share eigenkets), with

$$\hat{H}\,|+1\rangle = \frac{\hbar^2}{2\mu a^2}\,|+1\rangle .$$

Therefore the subsequent energy measurement is certain:

$$\boxed{E = \frac{\hbar^2}{2\mu a^2} \quad \text{with probability 1.}}$$

(e) First measure $E = \dfrac{9\hbar^2}{2\mu a^2}$, then measure L_z

The energy $E = \dfrac{9\hbar^2}{2\mu a^2}$ corresponds to $m^2 = 9$, i.e. the degenerate subspace spanned by $\{|-3\rangle, |+3\rangle\}$. Measuring this energy projects the state onto that subspace:

$$|\psi\rangle \;\longrightarrow\; |\psi_E\rangle \propto \frac{2}{5}\,|-3\rangle + \frac{3}{5}\,|+3\rangle .$$

The norm squared of the projected part is

$$\left|\frac{2}{5}\right|^2 + \left|\frac{3}{5}\right|^2 = \frac{4}{25} + \frac{9}{25} = \frac{13}{25},$$

so

$$\boxed{|\psi_E\rangle = \sqrt{\frac{4}{13}}\,|-3\rangle + \sqrt{\frac{9}{13}}\,|+3\rangle = \frac{2}{\sqrt{13}}\,|-3\rangle + \frac{3}{\sqrt{13}}\,|+3\rangle .}$$

Now measure L_z in this state. The possible outcomes are $L_z = \pm 3\hbar$ with probabilities given by the squared amplitudes:

$$\boxed{P(L_z = -3\hbar) = \left|\frac{2}{\sqrt{13}}\right|^2 = \frac{4}{13}, \qquad P(L_z = +3\hbar) = \left|\frac{3}{\sqrt{13}}\right|^2 = \frac{9}{13}.}$$

Problem 6: Particle on a Sphere

Solution

We are given the spherical harmonic (in position space)

$$|1,-1\rangle \;\widehat{=}\; Y_1^{-1}(\theta,\phi) = \sqrt{\frac{3}{8\pi}}\;\sin\theta\; e^{-i\phi}.$$

Recall the standard angular momentum eigenvalue equations,

$$\hat{L}_z Y_\ell^m = \hbar m\, Y_\ell^m, \qquad \hat{L}^2 Y_\ell^m = \hbar^2 \ell(\ell+1)\, Y_\ell^m.$$

(a) Act with L_z and find the eigenvalue

In spherical coordinates, the z–component of orbital angular momentum in position space is

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial\phi}.$$

Acting on $Y_1^{-1}(\theta,\phi)$:

$$\begin{aligned}
\hat{L}_z\, Y_1^{-1} &= -i\hbar \frac{\partial}{\partial\phi}\left(\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{-i\phi}\right) \\
&= -i\hbar\left(\sqrt{\frac{3}{8\pi}}\sin\theta\right)\left(-ie^{-i\phi}\right) \\
&= (-\hbar)\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{-i\phi} \\
&= -\hbar\, Y_1^{-1}(\theta,\phi).
\end{aligned}$$

Therefore,

$$\boxed{\hat{L}_z\, Y_1^{-1} = (-\hbar)\, Y_1^{-1}}$$

so the eigenvalue is $m\hbar$ with $m=-1$, as expected from $|\ell=1, m=-1\rangle$.

(b) Act with L^2 and find the eigenvalue

A defining property of spherical harmonics is

$$\hat{L}^2\, Y_\ell^m = \hbar^2 \ell(\ell+1)\, Y_\ell^m.$$

For $\ell = 1$ this gives

$$\boxed{\hat{L}^2\, Y_1^{-1} = \hbar^2(1)(2)\, Y_1^{-1} = 2\hbar^2\, Y_1^{-1}}.$$

So the eigenvalue of L^2 is $2\hbar^2$.

(c) Verify normalization on the unit sphere

Normalization over the surface of the unit sphere means

$$\int_0^{2\pi}\int_0^{\pi} |Y_1^{-1}(\theta,\phi)|^2 \, \sin\theta \, d\theta \, d\phi = 1, \qquad d\Omega = \sin\theta \, d\theta \, d\phi.$$

Compute the magnitude squared:

$$|Y_1^{-1}|^2 = \left(\sqrt{\frac{3}{8\pi}}\right)^2 \sin^2\theta \, |e^{-i\phi}|^2 = \frac{3}{8\pi}\sin^2\theta,$$

since $|e^{-i\phi}|^2 = 1$. Then

$$\begin{aligned}
\int |Y_1^{-1}|^2 \, d\Omega &= \int_0^{2\pi}\int_0^{\pi} \frac{3}{8\pi}\sin^2\theta \;\sin\theta \, d\theta \, d\phi \\
&= \frac{3}{8\pi}\left(\int_0^{2\pi} d\phi\right)\left(\int_0^{\pi} \sin^3\theta \, d\theta\right) \\
&= \frac{3}{8\pi}(2\pi)\left(\int_0^{\pi} \sin^3\theta \, d\theta\right) \\
&= \frac{3}{4}\left(\int_0^{\pi} \sin^3\theta \, d\theta\right).
\end{aligned}$$

Using the standard result

$$\int_0^{\pi} \sin^3\theta \, d\theta = \frac{4}{3},$$

we obtain

$$\frac{3}{4}\cdot\frac{4}{3} = 1.$$

Hence,

$$\boxed{\int_0^{2\pi}\int_0^{\pi} |Y_1^{-1}(\theta,\phi)|^2 \, \sin\theta \, d\theta \, d\phi = 1},$$

so the function is correctly normalized over the unit sphere.

(d) What can we infer from Fig. 7.19 about probability?

Figure 7.19 shows 3D polar plots of $|Y_\ell^m(\theta,\phi)|^2$, i.e. the *angular* probability density as a function of direction (θ,ϕ). For our state,

$$|Y_1^{-1}(\theta,\phi)|^2 = \frac{3}{8\pi}\sin^2\theta,$$

which is independent of ϕ. Therefore the probability is

- **maximal at the equator** ($\theta = \pi/2$), since $\sin^2(\pi/2) = 1$,
- **zero at the poles** ($\theta = 0, \pi$), since $\sin^2(0) = \sin^2(\pi) = 0$.

This matches the torus–like (donut) shape shown in Fig. 7.19 for $|Y_1^{\pm 1}|^2$: the particle is most likely to be found near the equatorial directions and least likely to be found along the $\pm z$ axis.

Problem 7: Spherical Harmonics Decomposition

Solution

A particle confined to the surface of a sphere is in the state

$$\psi(\theta,\phi) = \begin{cases} N\left(\dfrac{\pi^2}{4} - \theta^2\right), & 0 < \theta \le \dfrac{\pi}{2}, \\ 0, & \dfrac{\pi}{2} < \theta < \pi, \end{cases}$$

(where N is given in the problem). Since the particle is on the sphere, the inner product is

$$\langle f|g\rangle = \int_0^{2\pi}\int_0^{\pi} f^*(\theta,\phi)\, g(\theta,\phi)\, \sin\theta\, d\theta\, d\phi \equiv \int f^* g\, d\Omega.$$

We expand in spherical harmonics:

$$\psi(\theta,\phi) = \sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} c_{\ell m}\, Y_\ell^m(\theta,\phi), \qquad c_{\ell m} = \int \psi(\theta,\phi)\, Y_\ell^{m*}(\theta,\phi)\, d\Omega.$$

(a) Coefficients for $|0,0\rangle$, $|1,-1\rangle$, $|1,0\rangle$, $|1,1\rangle$

Because $\psi(\theta,\phi)$ is independent of ϕ, the ϕ–integral forces $m = 0$. In particular, for $m \neq 0$ the spherical harmonics contain $e^{im\phi}$, and

$$\int_0^{2\pi} e^{-im\phi}\, d\phi = 0 \quad (m \neq 0),$$

so

$$\boxed{c_{1,-1} = 0, \qquad c_{1,1} = 0.}$$

We now compute c_{00} and c_{10}. Using

$$Y_0^0 = \frac{1}{\sqrt{4\pi}}, \qquad Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta,$$

and the fact that ψ is nonzero only on $0 \le \theta \le \pi/2$:

Coefficient c_{00}.

$$\begin{aligned} c_{00} &= \int \psi\, Y_0^{0*}\, d\Omega \\ &= \int_0^{2\pi}\int_0^{\pi/2} N\left(\frac{\pi^2}{4} - \theta^2\right)\left(\frac{1}{\sqrt{4\pi}}\right)\sin\theta\, d\theta\, d\phi \\ &= \frac{2\pi N}{\sqrt{4\pi}}\int_0^{\pi/2}\left(\frac{\pi^2}{4} - \theta^2\right)\sin\theta\, d\theta. \end{aligned}$$

The integral evaluates to

$$\int_0^{\pi/2}\left(\frac{\pi^2}{4} - \theta^2\right)\sin\theta\, d\theta = -\pi + 2 + \frac{\pi^2}{4}.$$

Also $\dfrac{2\pi}{\sqrt{4\pi}} = \sqrt{\pi}$, so

$$\boxed{c_{00} = N\sqrt{\pi}\left(-\pi + 2 + \frac{\pi^2}{4}\right).}$$

Coefficient c_{10}.

$$\begin{aligned} c_{10} &= \int \psi\, Y_1^{0*}\, d\Omega \\ &= \int_0^{2\pi}\int_0^{\pi/2} N\left(\frac{\pi^2}{4} - \theta^2\right)\left(\sqrt{\frac{3}{4\pi}}\cos\theta\right)\sin\theta\, d\theta\, d\phi \\ &= 2\pi N\sqrt{\frac{3}{4\pi}}\int_0^{\pi/2}\left(\frac{\pi^2}{4} - \theta^2\right)\sin\theta\cos\theta\, d\theta. \end{aligned}$$

The integral evaluates to

$$\int_0^{\pi/2}\left(\frac{\pi^2}{4} - \theta^2\right)\sin\theta\cos\theta\, d\theta = \frac{1}{4} + \frac{\pi^2}{16} = \frac{4+\pi^2}{16}.$$

Also $2\pi\sqrt{\dfrac{3}{4\pi}} = \sqrt{3\pi}$, so

$$\boxed{c_{10} = N\sqrt{3\pi}\,\frac{4+\pi^2}{16}.}$$

Collecting the four requested coefficients:

$$\boxed{c_{00} = N\sqrt{\pi}\left(-\pi + 2 + \frac{\pi^2}{4}\right), \quad c_{1,-1} = 0, \quad c_{10} = N\sqrt{3\pi}\,\frac{4+\pi^2}{16}, \quad c_{1,1} = 0.}$$

(b) Probabilities for a measurement of L^2

On a sphere, the spherical harmonics are eigenstates of L^2:

$$\hat{L}^2 Y_\ell^m = \hbar^2\ell(\ell+1)Y_\ell^m.$$

Thus:

$$\ell = 0 \Rightarrow L^2 = 0, \qquad \ell = 1 \Rightarrow L^2 = 2\hbar^2.$$

Using only the coefficients found in (a),

$$\boxed{P(L^2 = 0) = |c_{00}|^2, \qquad P(L^2 = 2\hbar^2) = \sum_{m=-1}^{1}|c_{1m}|^2 = |c_{10}|^2.}$$

Explicitly,

$$\boxed{P(L^2 = 0) = N^2\pi\left(-\pi + 2 + \frac{\pi^2}{4}\right)^2, \qquad P(L^2 = 2\hbar^2) = N^2\, 3\pi\left(\frac{4+\pi^2}{16}\right)^2.}$$

(Note: because ψ is not exactly a linear combination of only $\ell = 0, 1$ harmonics, the remaining probability lies in $\ell \geq 2$ terms that are being neglected in this approximation.)

(c) Approximate time-dependent wavefunction using the terms in (a)

The Hamiltonian is given by

$$H_{\text{sphere}} Y_\ell^m(\theta,\phi) = \frac{\hbar^2}{2I}\,\ell(\ell+1)\,Y_\ell^m(\theta,\phi),$$

so the energy eigenvalues are

$$E_\ell = \frac{\hbar^2}{2I}\ell(\ell+1).$$

Therefore

$$E_0 = 0, \qquad E_1 = \frac{\hbar^2}{2I}\cdot 2 = \frac{\hbar^2}{I}.$$

Keeping only the $\ell = 0$ and $\ell = 1$ pieces found in (a), the approximate time-dependent wavefunction is

$$\boxed{\psi(\theta,\phi,t) \approx c_{00}\,Y_0^0(\theta,\phi)\,e^{-iE_0 t/\hbar} + c_{10}\,Y_1^0(\theta,\phi)\,e^{-iE_1 t/\hbar}}$$

i.e.

$$\boxed{\psi(\theta,\phi,t) \approx c_{00}\,Y_0^0(\theta,\phi) + c_{10}\,Y_1^0(\theta,\phi)\,e^{-i(\hbar/I)t},}$$

with c_{00} and c_{10} given above.